

ADARSH SHETH

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OBJECTIVE

Rising Senior in Aerospace Engineering with hands-on experience with CAD, CFD, mechanical assembly, mission analysis, and systems integration experience in design team and competition settings, seeking a Summer 2027 internship in any location, specializing in structures, mechanical engineering, product design, or traditionally aerospace roles.

EDUCATION

Texas A&M University, College Station, TX

Expected May 2028

Bachelor of Science in Aerospace Engineering

- Coursework: Structural Analysis II, Materials Science, Mechanics of Materials, Fluid Dynamics, Control Theory, Vehicle Dynamics, Orbital Mechanics, Thermodynamics, Numerical Simulation, Circuits, Program Design, High Speed Aerodynamics
- Leadership: program coordinator for UpLift (HS drone design program), design team lead for SEDS Lunar Search & Rescue

EXPERIENCE

Systems Engineer Intern

Jun 2026 – Aug 2026

Raytheon

Woburn, MA

- Architected satellite acquisition pipeline across 15K+ LEO parse (ECI/ECEF/AER) under FOV/kinematic/RSO signature constraints for Over-the-Horizon Radar (OTHR) modernization initiative, slashing runtime by 50 minutes across benchmarks
- Developed standalone target-tracking script (MATLAB, Bitbucket) with dynamic geospatial trajectory overlay and partner-supplied propagator integration for live customer demonstration, easing radar operations and eliminating external dependencies
- Automated TLE fetch (Space-Track API) and refactored data schemas to Alpha-5 to ensure downstream propagation continuity

Aerodynamics Engineer

Sep 2025 - Present

Texas A&M Vertical Flight Design Team

College Station, TX

- Executed 100+ parametric CFD analysis (SimScale) for 25-lb hexacopter, feeding forces data into 6DoF dynamics model, ensuring 100% flight stability under 15-kt gusts and autonomous payload pick-up and delivery precision with 3-by-3 ft zone
- Integrated translational and hover state mass-power functions into 1000 flight paths (Monte Carlo) to optimize firefighting response, achieving -27% time and +17% points increase, winning 1st presentation and 2nd technical report at DBVF competition
- Iterated arm-folding mechanism and authored field installation guide for compact transport and 50% faster pre-flight deployment

Fluid Systems Engineer

Feb 2025 – Jan 2026

Texas A&M Rocket Engine Design Team

College Station, TX

- Developed N2O leak/pressure test (P&ID, procedure), Upper Nitrous FMEA, fluid systems assembly schematics, and 11' tank manufacturing strategy for liquid bi-propellant engine Elysium 2.0 development, presenting at and clearing PDR/CDR milestones
- Automated BoM with cost authentication, part overrides, and dynamic inventory for hardware traceability and financial records

Lead Aircraft Sizing & Performance Engineer

Nov 2025

Aggies Ignite Challenge

College Station, TX

- Designed lift+cruise S&R eVTOL for extreme operations (10K ft altitude, 40 kt winds, 50% visibility), winning 1st place
- Developed iterative conceptual aircraft sizing algorithm (Python) to evaluate 2M+ configurations against conflicting aerodynamic and weight constraints, optimizing planform area and rotor fractions to cut MTOW by 112-lbs, sustaining cruise speed of 42 mph
- Validated with sourcing feasibility analysis, built subsystem CAD model (Fusion 360), and sliced for 3D printing (Bambu Studio)

Co-Founder, Captain, Mechanical Engineer

Mar 2022 - Jun 2024

Dougherty Valley VEX Robotics

San Ramon, CA

- Oversaw CAD (Fusion 360), T&E, machining, mechanical assembly, strategy optimization, documentation, and systems integration for 12-member team, winning 8 awards, 2 world championship qualifiers, and ranking in top 2% of 20,000 teams
- Established chain of command, project management workflow, underclassmen mentorship system, and \$9,000 in funding from the ground up, outperforming 4 sister teams and setting a historical achievement of 100% world qualification rate
- Authored 1 technical, 1 management, and 2 operational guides, creating documentation for programs with \$30K+ annual revenue

PROJECTS

- **Aggies INVENT**: Researched, designed, prototyped, and presented honeycomb plate-type SMR to maximize heat transfer
- **IGNITE Design Challenge**: rocket design, aerodynamics, landing simulation (Python), and presentation scoring 95/100

AWARDS & SKILLS

- Eagle Scout & 4 Palms, 1st Place Aggies Ignite Challenge, VEX Worlds 2023 & 2024, 8x Robotics Awards, 3X PVSA, CSF
- Autodesk Fusion 360, SolidWorks, Onshape CAD, DFA, Kinematic Analysis, Assembly, FMEA, GD&T, P&ID, T&E, V&V, SimScale CFD, OpenFOAM, XFLR5, MATLAB, Simulink, Propeller/Rotor Analysis, Aerodynamic Sizing, Pneumatics, 6DOF, MATLAB, Python, C++, VBA, Git, GitHub/Bitbucket, Monte Carlo Simulation, Numerical Methods, Confluence